

Backyard Bird Identification Using Computer Vision Tier 2 Project

ITAI 1378 – 12322

Group 8

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The Problem

- Bird populations are a key indicator of ecosystem health, and researchers have tracked them since 1900 (Audubon Christmas Count).
- Citizen science, like Cornell's Merlin Bird ID, now drives most local bird data collection.
- Our goal: extend that effort with an autonomous camera system that continuously observes and records bird species at a single location, exporting the data for the research community.



What We Built

- Our application uses an uploaded video as input, processed by YOLO11Large to detect birds in the image, and then uses a custom trained classifier model to identify between 7 bird species, and outputs an image and metadata of the highest confidence tracked bird for cataloging.



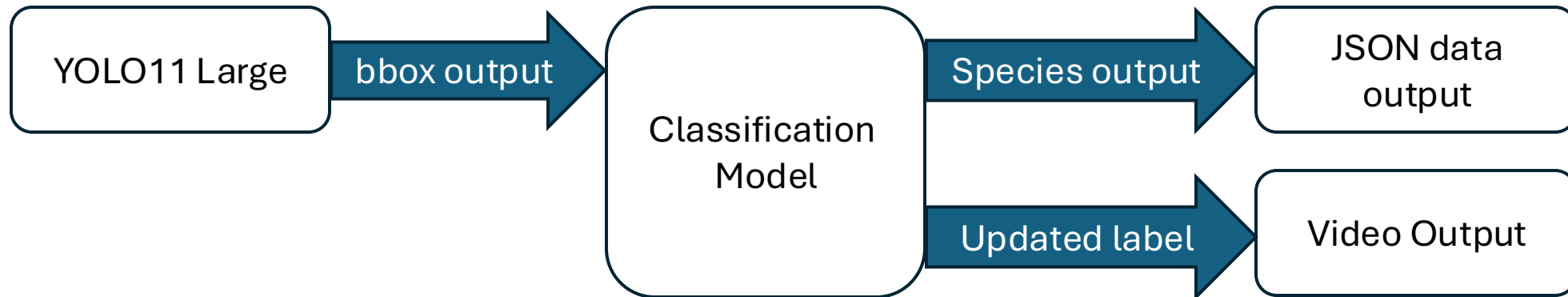
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    "track_id": 3,
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  }
]
```


Demo



How It Works

- Working in the PyTorch framework, we use the YOLO11 Large model for detecting birds in the video frame
- The output of the detection model is fed into the custom classification model and detects bird species



Dataset's Used

1. General bird detection (Stage 1 YOLO11 Large):

Used pretrained as is, no custom training data required YOLO11 already detects the COCO "bird" class out of the box.

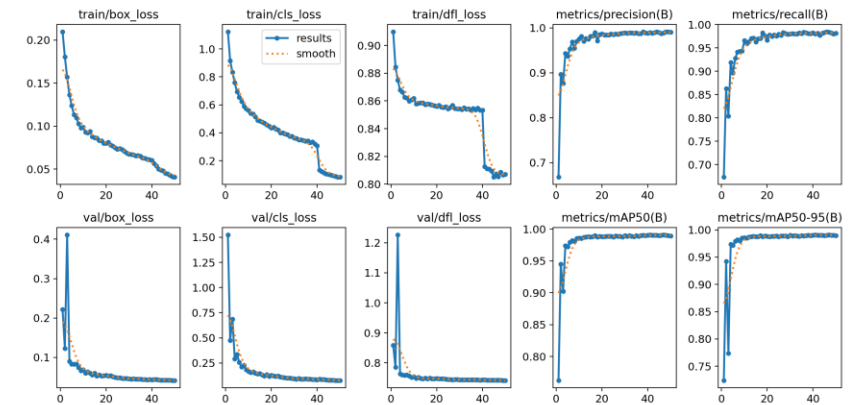
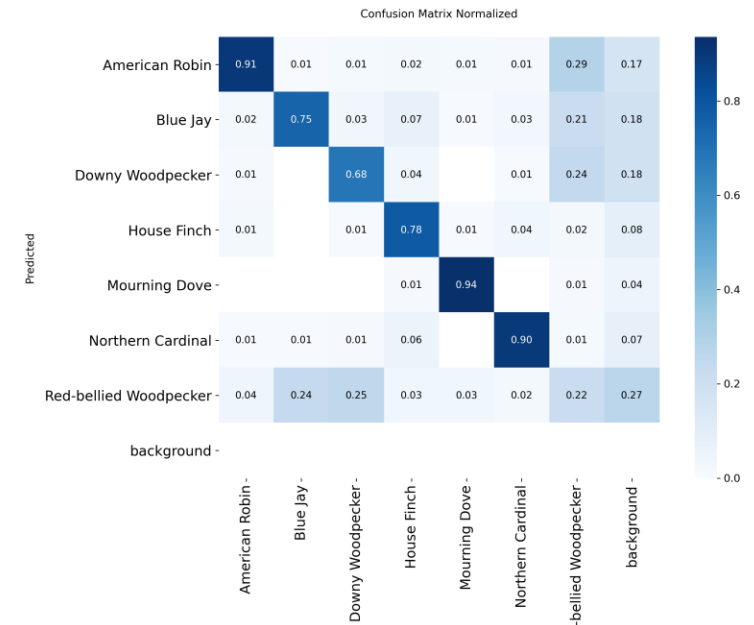
2. Species classification (Stage 2 custom multi-bird model):

Trained on images scraped from Pixabay, one search per target species 7 species total: American Robin, Blue Jay, Downy Woodpecker, House Finch, Mourning Dove, Northern Cardinal, Red-bellied Woodpecker.

Source	Role in Final Build	Size / Labels
Pretrained YOLO11 (COCO)	General bird detection	Pretrained no custom data needed
Pixabay (7 species searches)	Species classifier training data	~500 images/species, ~3,500 total, labeled by species
Kaggle Bird Detection (gpiosenka)	Reference only informed bounding-box annotation format during early experimentation	~2,000 images, not used as final training data

Target Results vs Results

- A target of 90% initial detection accuracy and 80% classification accuracy. We lowered the initial detection to 12% for YOLO to initially detect, and the classification model achieved 99% accuracy in its training, running at a threshold of 45% confidence when running in the pipeline.
- To detect each bird in under 150ms/frame was the target for achieving livestream use. We got under 40ms.



What Changed From the Blueprint

Updated the deliverables from a
livestream to a recorded video

Updated the dataset access to
remove the paid dataset

Attempted further tracking of the
birds in the video clips to get a
better output for the data
collection



Challenges and Fixes

- Training of a custom model to outperform YOLO11Large on detection
Fix - We used YOLO as the detector
- Properly using Colab system resources – moving to a GPU runtime got us under 40ms, within the requirements for a realtime pipeline
- Scoped custom dataset to 7 birds to handle data size

Lessons and Next Steps

Lessons learned:

- Two-stage pipelines compound errors, and cross-frame tracking helped smooth this out
- Planned datasets - LaSBIRD wasn't allowed - should have backup datasets like the Pixabay ones we used.
- Batch video was the right scope call over live RTSP for a working demo
- Testing 6 models head-to-head beat guessing based on "newest"

Next steps:

- Connect the pipeline to a live RTSP stream
- Grow the training set with more images and species
- Build a proper hand-labeled test set for accuracy measurement